

Email: jnie47@gatech.edu

Research interests My research interests lie mainly in extremal combinatorics and probabilistic combinatorics, but I keep my brain open for any interesting math problems.

Positions

Visiting Assistant Professor,	Georgia Institute of Technology
Supervisor: Xiaoyu He	2024 August – present

Postdoctoral Researcher,	Shanghai Center for Mathematical Sciences
Supervisor: Hehui Wu	2022 September – 2024 July

Short-term postdoc,	Max Planck Institute for Mathematics in the Sciences
Supervisor: Raffaella Mulas	2022 May-August

Education

Ph.D. in Mathematics, University of California, San Diego	2016 – 2022
Advisor: Jacques Verstraëte. <i>GPA: 4.00/4.00</i>	

Exchange Semester, University of California, Berkeley	2014 Fall
<i>GPA: 4.00/4.00</i>	

B.S. in Mathematics, Nankai University	2012 – 2016
<i>GPA: 90.41/100</i>	

High School, Shenzhen Middle School	2009 – 2012
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Papers (ordered by submission time)

29. Matching and intersection problems for non-trivial r -partite r -uniform hypergraphs (with P. Frankl)
Submitted. [arxiv](#).

28. Hypergraph Ramsey numbers with quasipolynomial growth rate (with X. He, L. Post, J. Verstraëte)
Submitted. [arxiv](#).

27. New Sidorenko-type inequalities in tournaments (With X. He, N. Mani, N. Tung and F. Wei)
Submitted. [arxiv](#).

26. Decomposition of Cliques into k-Star-Forests (With Y. Ren and H. Wu)
Submitted. [arxiv](#).

25. Maximum in-general-position set in a random subset of $\mathbb{F}_q^d$ (With Y. Chen, J. Yu and W. Zhang)

To appear on *Journal of Combinatorial Theory, Series B*. [arxiv](#).

24. Evasive sets, twisted varieties, and container-clique trees (With J. Lim and J. Zeng)

SODA 2026. [arxiv](#).

23. Off-Diagonal Ramsey Numbers for Linear Hypergraphs (With X. He, Y. Wigderson and H. Yu)

Combinatorics, Probability and Computing (2026). [arxiv](#).

22. Supersaturation of odd linear cycles (With L. Deng, J. Han and S. Spiro)

To appear on *Electronic Journal of Combinatorics*. [arxiv](#).

21. Generalized Erdős-Rogers problems for hypergraphs (With X. He)

European Journal of Combinatorics (2026). [arxiv](#).

20. On tight tree-complete hypergraph Ramsey numbers

Journal of Graph Theory (2026). [arxiv](#).

19. Random Turán Problems for $K_{s,t}$ Expansions (With S. Spiro)

To appear on *SIAM Journal on Discrete Mathematics*. [arxiv](#).

18. On the Matching Problem in Random Hypergraphs (With P. Frankl and J. Wang)

Discrete Mathematics (2026). [arxiv](#).

17. Off-diagonal Ramsey number for slowly growing hypergraphs (with S. Mattheus, D. Mubayi and J. Verstraëte)

Random Structures & Algorithms (2025). [arxiv](#).

16. On odd covers of cliques and disjoint unions. (with C. Buchanan, A. Clifton, E. Culver, P. Frankl, K. Ozeki, P. Rombach, and M. Yin)

To appear on *Journal of Graph Theory*. [arxiv](#).

15. Random Turán Problems for Hypergraph Expansions. (With S. Spiro)

Submitted. [arxiv](#).

14. Sidorenko Hypergraphs and Random Turán Numbers. (With S. Spiro)

Combinatorial Theory (2025). [arxiv](#).

13. Random Turán and counting results for general position sets over finite fields. (with Y. Chen, X. Liu, and J. Zeng)

SCIENCE CHINA Mathematics (2025). [arxiv](#).

12. Random Turán theorem for expansions of spanning subgraphs of tight trees.

SIAM Journal on Discrete Mathematics (2026). [arxiv](#).

11. Turán theorems for even cycles in random hypergraph.

Journal of Combinatorial Theory, Series B (2024). [arxiv](#).

10. The Number of Edge Colorings with Small Independence Number and No Monochromatic H . (with X. Hu and Q. Lin)

The Electronic Journal of Combinatorics (2025). [arxiv](#).

9. On asymptotic local Turán problems. (with P. Frankl)

The Moscow Journal of Combinatorics and Number Theory (2023). [arxiv](#).

8. On asymptotic packing of convex geometric and ordered graphs. (with E. Surya, and J. Zeng)

Journal of Graph Theory (2023). [arxiv](#).

7. Odd Covers of Graphs. (with C. Buchanan, A. Clifton, E. Culver, J. O'Neill, P. Rombach, and M. Yin)

Journal of Graph Theory (2023). [arxiv](#).

6. On Turán numbers for disconnected hypergraphs. (with R. Mulas)

Acta Mathematica Hungarica 170, 168–182 (2023). [arxiv](#).

5. On Asymptotic Packing of Geometric Graphs. (with D. Cranston, J. Verstraëte, and A. Wesolek)

Discrete Applied Mathematics 322 (2022). [arxiv](#).

4. Maximal independent sets in clique-free graphs. (with X. He, and S. Spiro)

European Journal of Combinatorics (2022). [arxiv](#).

3. Ramsey Numbers for Non-trivial Berge Cycles. (with J. Verstraëte)

SIAM Journal on Discrete Mathematics (2022). [arxiv](#).

2. Triangle-free Subgraphs of Hypergraphs. (with S. Spiro, and J. Verstraëte)

Graphs and Combinatorics (2021). [arxiv](#).

1. Randomized greedy algorithm for independent sets in regular uniform hypergraphs with large girth. (with J. Verstraëte)

Random Structures & Algorithms (2021). [arxiv](#).

Teaching

As Instructor

at Georgia Institute of Technology:

3406 - A second course on Linear Algebra Fall 2025

4032 - Combinatorial Analysis Spring 2025/2026

3012 - Applied Combinatorics Fall 2024

at Fudan University:

Graph Theory Fall 2023

at UCSD

MATH 20C - Calculus&Analyt Geom/Sci&Engnr Summer I 2021

As Teaching Assistant at UCSD

MATH 154 - Discrete Math & Graph Theory Winter 2022

MATH 154 - Discrete Math & Graph Theory Winter 2021

MATH 154 - Discrete Math & Graph Theory Fall 2020

MATH 20E - Vector Calculus Summer II 2020

MATH 154 - Discrete Math & Graph Theory Spring 2020

MATH 154 - Discrete Math & Graph Theory Winter 2020

MATH 170A - Intro Numerical Analys/Linear Fall 2019

MATH 220C - Complex Analysis Spring 2019

MATH 18 - Linear Algebra Spring 2019

MATH 170B - Intro/Numerical Analy/Approxim Winter 2019

MATH 170A - Intro Numerical Analys/Linear Fall 2018

MATH 142A - Introduction to Analysis I Summer I 2018

MATH 184A - Combinatorics Spring 2018

MATH 184A - Combinatorics Winter 2018

MATH 184A - Combinatorics Fall 2017

MATH 20B - Calculus/Science & Engineering Winter 2017

MATH 20A - Calculus/Science & Engineering Fall 2016

Service

Co-orgarnizing the [GaTech Combinatorics Seminar](#)

Co-orgarnizing the [SCMS Combinatorics Seminar](#)

Served as referee for the following journals:

- Journal of London Mathematical Society
- Forum of Mathematics Sigma
- Journal of Combinatorial Theory Series B
- Combinatorica
- Random Structures & Algorithms
- Combinatorics, Probability and Computing
- SIAM Journal of Discrete Mathematics

- Journal of Graph Theory
- European Journal of combinatorics
- Discrete Applied Mathematics
- Discrete Mathematics
- Electronic Journal of combinatorics
- Graphs and Combinatorics
- Australasian Journal of Combinatorics
- Journal of Combinatorial Optimization